

MAFABI HUSSEIN

Software Engineer | Backend & Full-Stack Development

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PROFESSIONAL SUMMARY

Software Engineer with a BSc in Computer Science and 4+ years of hands-on experience building and deploying software across web, backend, mobile, and AI-enabled applications. I specialize in backend engineering while working across the full stack, with experience taking software from requirements and system design through frontend development, API implementation, database design, deployment, and troubleshooting. I have built multi-application commerce systems, organizational web platforms, mobile computer-vision applications, and supporting APIs. My work includes authentication and authorization, relational data modelling, business workflows, third-party integrations, content management, deployment, and debugging. Although I am early in my professional career as a graduate, I am not new to software development. I have spent several years building, breaking, debugging, deploying, and improving real applications and am seeking a junior Software Engineering role where I can contribute while continuing to grow with an experienced engineering team.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, Kotlin, HTML, CSS

Backend: Node.js, Express, NestJS, FastAPI, Django, REST API development, authentication, authorization, business logic

Frontend: React, Next.js, Vue.js, Tailwind CSS, Progressive Web Apps, responsive web development

Databases: PostgreSQL, MySQL, MongoDB, Prisma, Supabase, relational database design, transactions and data consistency

Mobile and AI: Android, TensorFlow Lite, computer vision, image recognition, AI and third-party API integration

Development and Deployment: Git, GitHub, Vercel, Render, Railway, environment configuration, application deployment, production debugging

EXPERIENCE

SOFTWARE ENGINEER / FREELANCE DEVELOPER

Self-Employed | 2022 – Present | Uganda

- Build and deliver software applications from requirements through implementation, deployment, troubleshooting, and ongoing improvement.
- Translate business and user requirements into application structure, database models, API contracts, and functional interfaces.
- Develop backend services and APIs covering authentication, authorization, validation, business logic, and data persistence.
- Design and implement database structures for application workflows involving users, roles, products, orders, content, and operational data.
- Develop responsive web applications and Progressive Web Apps using React, Next.js, TypeScript, and JavaScript, with backend services built using Node.js or Python.
- Deploy applications using Vercel, Render, and Railway, including environment configuration, release management, and production troubleshooting.
- Investigate and resolve problems across frontend applications, APIs, databases, integrations, and deployment environments, including issues that emerge under concurrent usage.
- Work independently across the software development lifecycle and make practical architecture and technology decisions based on project requirements and constraints.

DATA ANNOTATOR

Samasource Uganda Ltd | Jul 2021 – Nov 2021 | Gulu, Uganda

- Annotated and validated datasets used in computer vision and natural language processing workflows.
- Maintained project-specific quality standards while processing high-volume annotation tasks.
- Developed practical understanding of dataset quality, annotation consistency, and requirements of machine-learning pipelines.

SELECTED WORK

A selection of software systems and public platforms I have designed, developed, and deployed. Additional work is available through my portfolio.

OAKLEY COMMERCE PLATFORM

Sole Developer & System Architect | *React, TypeScript, Node.js, Express, PostgreSQL, Supabase*

- Designed and developed the customer storefront, vendor operations application, backend API, and supporting database structure.
- Implemented product, cart, order, authentication, authorization, and vendor-management workflows.
- Designed relational data structures and application boundaries to keep business operations consistent across the platform.
- Investigated and resolved issues involving concurrent operations, data consistency, payment workflows, and deployment.
- Deployed and maintained the system while troubleshooting issues across the frontend, backend, database, and hosting environment.

ACHOLI WOMEN IN HEALTH FOUNDATION (AWIHF)

Software Developer | *Next.js, TypeScript, Sanity CMS, Neon PostgreSQL, Prisma, Cloudinary, Resend*

- Designed and developed the organization's public web platform around its programmes, impact reporting, news, health communication, and public information needs.
- Integrated Sanity CMS and PostgreSQL-backed data workflows to support structured content and future expansion.
- Built responsive interfaces and reusable components using Next.js and TypeScript.
- Integrated supporting services including Cloudinary and Resend for content and communication workflows.
- Deployed the platform for public use and structured it so the organization can continue managing and expanding its content.

OBJECTID | IMAGE IDENTIFICATION SYSTEM

Primary Developer | *Kotlin, Android, TensorFlow Lite, Python, Imagga API*

- Built the Android application and recognition workflow, including camera/gallery input, image selection, region annotation, processing, and result presentation.
- Integrated lightweight TensorFlow Lite inference for on-device recognition and an external computer-vision API for broader recognition capability.
- Worked across the mobile application, inference pipeline, API integration, and result presentation.
- Designed the system around practical constraints including mobile hardware limitations, network availability, and model capability.

EDUCATION

BSc Computer Science

Kampala International University | 2022 – 2025 | Uganda

Higher Education Access Certificate

Gulu University | 2020 – 2021 | Uganda

CERTIFICATIONS

Database Management Systems | Great Learning Academy

Data Science Foundations | Great Learning Academy

ADDITIONAL INFORMATION

Available full-time | Open to Software Engineering, Backend Engineering, and Full-Stack opportunities | Portfolio: mhussein.vercel.app